Quantifying Coupling Errors in Atmosphere-Ice-Ocean Coupling

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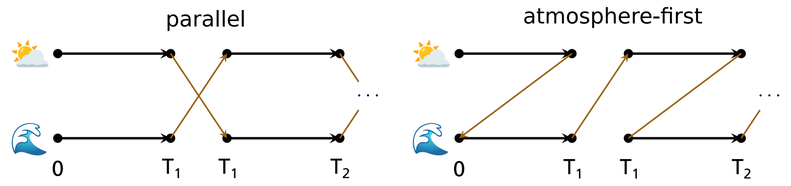
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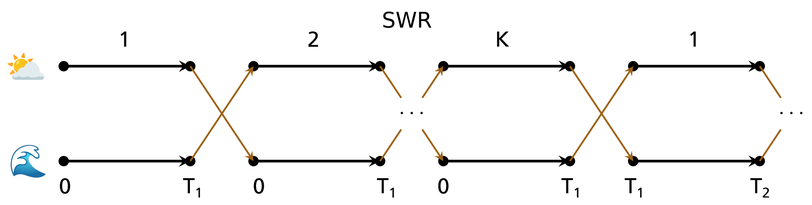


In **Earth system models**, the atmosphere, ocean, and sea ice are coupled at the sea surface using boundary conditions. In other words, they solve a **coupled problem**. We use an **iterative coupling algorithm** to study numerical coupling errors in the **EC-Earth** single column model (**AOSCM**).

2 Method: Schwarz Waveform Relaxation (SWR)

A simulation is divided into **coupling windows**. After each coupling window, the models exchange time-averaged data.





We interpret this as **one step** of Schwarz Waveform Relaxation, an **iterative** algorithm. Not iterating is computationally cheap, but leads to a numerical coupling error.

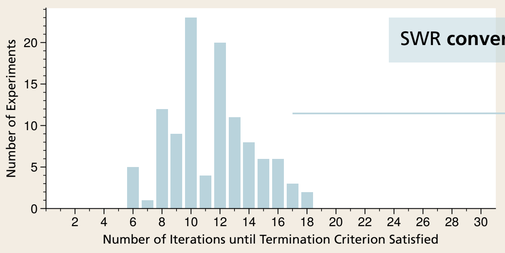
The iteration will converge if the coupled problem is set up correctly. We check whether it converges in practice. If yes, we compute the coupling error:

$$e_{\text{atm-first}} = \|u_{\text{atm-first}} - u_K\|$$

3 Case Study: EC-EARTH AOSCM [1]

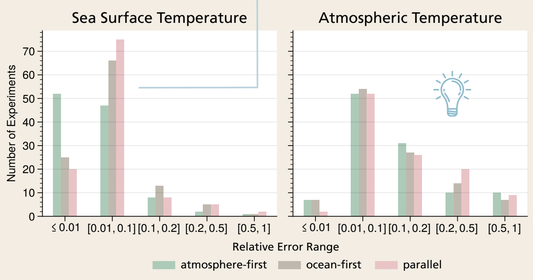
Ice-free conditions

PAPA station, July 2014, 112 two-day forecasts



SWR **converges** consistently.

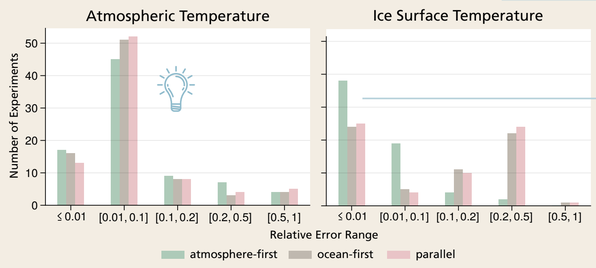
Atmosphere-first best for **ocean variables**, since solar radiation phase error dominates.

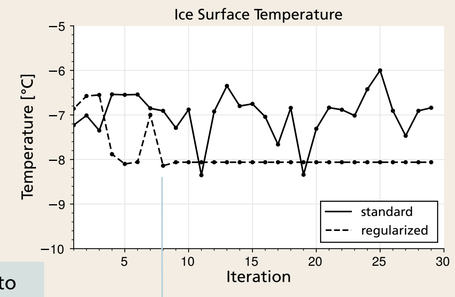


Ice-covered sea surface

MOSAiC, April 2020, 84 two-day forecasts

Similar pattern as for ice-free experiments.



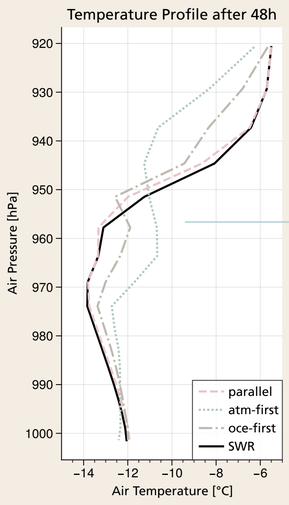


SWR **non-convergence** due to **discontinuous** albedo physics. Regularizing the albedo scheme **restores** convergence.

For all experiments

No algorithm is a clear winner for atmospheric variables!

Large ice coupling errors.



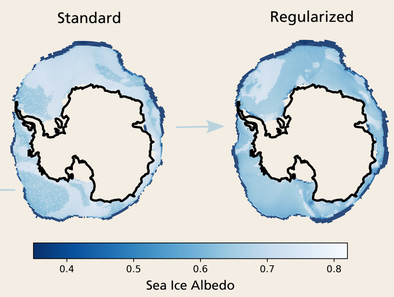
$e_{max}(T^{ice})$	**4.66** °C
e_{max}(SST)	0.58 °C
$e_{max}(T^{A})$	**7.48** °C

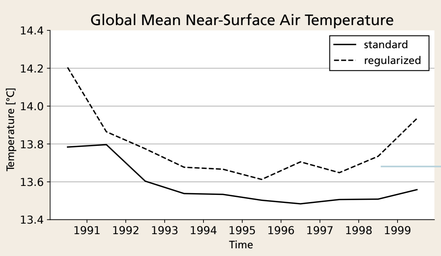
Large coupling errors in atmosphere linked to **discontinuous** convection scheme.

4 Learning from the Case Study

Sensitivity experiment in EC-Earth 4

Regularizing the albedo parameterization **removes** an artificial, **noisy pattern** in 3D.



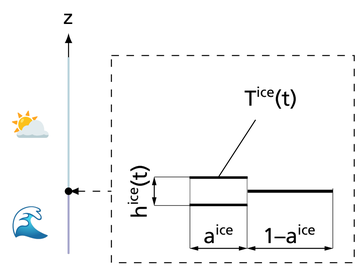


The regularization affects global mean temperature.

Atmosphere-ice analysis

Atmosphere-ocean and atmosphere-ice coupling are governed by **different** physical processes.

We present and study a simple model for **combined atmosphere-ice-ocean** SWR convergence analysis in [2]:



Conclusions & References

SWR is a **tool** for model **development**:
→ Numerical coupling errors can be large.
→ Identify issues in physics parameterizations.

SWR is a **tool** for model **understanding**:
→ Which processes govern convergence of the iteration?
→ How does this affect the coupling error?

[1] V. Schüller, F. Lemarié, P. Birken, E.Blayo (2025): Quantifying coupling errors in atmosphere-ocean-sea ice models, Geosci. Model Dev.

[2] V. Schüller, P. Birken, H. Kjellson (in press). Analysis of Bulk Interface Conditions for Atmosphere-Ice-Ocean Coupling: DD29 Proceedings.

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